Roberto Giménez Conejero

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Education

Ph.D in Mathematics 2017–2021
Thesis: *Singularities of germs and vanishing homology*
Advisor: J. J. Nuño-Ballesteros
Universitat de València, Valencia, Spain

M.Sc. in Mathematics and Applications 2016–2017
Thesis: *Vanishing cycles for analytic maps.*
External advisor: J. J. Nuño-Ballesteros. Internal tutor: O. Villamayor
Universidad Autónoma de Madrid, Madrid, Spain

B.Sc. in Mathematics 2012–2016
Thesis: *Maps from the plane to the plane and a theorem of Haefliger.*
Advisor: R. Oset Sinha
Universitat de València, Valencia, Spain

Distinctions, Grants and Positions

Assistant Professor 2026–now
Universidad Politécnica de Madrid

Assistant Professor 2025–2026
Universidad Autónoma de Madrid & ICMAT

Researcher Fellowship 2024
Mid Sweden University

Vicent Caselles prize 2023
Spanish Royal Society of Mathematics (RSME) and *BBVA bank*

Rényi Postdoctoral Fellowship 2021–2024
Alfréd Rényi Institute of Mathematics

Predoctoral grant 2017–2021
Ministerio de Ciencia, Innovación y Universidades Grant FPU16/03844 in Universitat de València

M.Sc. Grant 2016–2017
of the Department of Mathematics of *Universidad Autónoma de Madrid*

Bachelor's Degree Excellence Award 2016
Universitat de València

Overall classification of the Bachelor's Degree 2016
9.76/10 and 3.76/4

B.Sc. Grant 2012–2016
Ministerio de Educación y Formación Profesional General Grants (4 in total)

Collaboration Internship Grant 2015–2016
Ministerio de Educación y Formación Profesional

Severo-Ochoa-ICMAT grant: Introduction to research 2016
Instituto de Ciencias Matemáticas, Madrid

Publications

The image Milnor number and excellent unfoldings (w/ J.J. Nuño-Ballesteros),
Q. J. Math., 2022, DOI: 10.1093/qmath/haab019

Singularities of mappings on ICIS and applications to Whitney equisingularity (w/ J.J. Nuño-Ballesteros),
Adv. Math., 2022, DOI: 10.1016/j.aim.2022.108660

Thom condition and monodromy (w/ Lê Dũng Tráng and J.J. Nuño-Ballesteros),
Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM, 2022, DOI: 10.1007/s13398-022-01353-y

A weak version of the Mond conjecture (w/ J.J. Nuño-Ballesteros),
Collect. Math., 2023, DOI: 10.1007/s13348-023-00411-x

A note on complex plane curve singularities up to diffeomorphism and their rigidity (w/ A. Fernández-Hernández),
Res. Math. Sci., 2024, DOI: 10.1007/s40687-024-00439-w

Isotypical components of the homology of ICIS and images of deformations of map germs,
Proc. Roy. Soc. Edinburgh Sect. A Math., 2025, DOI: 10.1017/prm.2025.8

Good real pictures of corank one map germs from the n-space to the $(n+1)$-space (w/ I. Breva Ribes),
Res. Math. Sci., 2026, DOI: 10.1007/s40687-025-00577-9

Signature of the Milnor fiber of parametrized surfaces (w/ G. Pintér),
J. Lond. Math. Soc., 2026, DOI: 10.1112/jlms.70699

Sent manuscripts

Isotypes of ICIS and images of deformations of map germs,
arXiv:2207.05196

Embedded topological triviality of separable families of singularities (w/ A. Lind and A. Menegon),
arXiv:2511.23185

Presentations

Courses

Singularities of maps and immersion theory (6h) 2024
Iberosing International Workshop, Universidad Politécnica de Madrid

Curve singularities and their deformations (TA session) 2023
Winter school in singularities and low-dimensional topology, Rényi Institute

Projective geometry (in Spanish) (videos) 2020
Summer school SEMF 2020, On-line

Introduction to topology (in Spanish) 2016
SEMF and Faculty of Physics course, *Univeristat de València*

Simplicial and singular homology (in Spanish), 20h 2015–2016
Extracurricular seminar, Department of Geometry and Topology, *Universitat de València*

Invited talks

***A*-finite monomial maps are very rare** 2026
International School on Singularity Theory at Mid Sweden University, Sundsvall, Sweden

Good real pictures of complex maps 2025

13th Mini Workshop on Singularities, Geometry and Differential Equations, São Carlos, Brazil

Good real pictures of complex maps 2024
18th International Workshop on Real and Complex Singularities, Valencia, Spain

Complex curve singularities modulo smooth diffeomorphism 2023
Spanish Topology Meeting, Alicante, Spain

Halfway towards the Mond conjecture 2023
The singular workshop, Budapest, Hungary

Computing signatures of nearby fibres of parametrized complex surfaces 2023
VI Congreso de jóvenes investigadores de la RSME (Topology session), León, Spain

Computing a difficult invariant of holomorphic map germs 2023
VI Congreso de jóvenes investigadores de la RSME (Singularities session), León, Spain

Towards Mond's conjecture 2022
17th International Workshop on Real and Complex Singularities, São Carlos, Brazil

Hazañas y enigmas de los gérmenes de aplicaciones holomorfas (in Spanish) 2021
Special hybrid workshop for PhD students in singularities, Madrid, Spain

Contributed and seminar talks

Complex curve singularities modulo smooth diffeomorphism 2023
7th International Workshop on Singularities in Geometry and applications, Fortaleza, Brazil

Complex curve singularities modulo smooth diffeomorphism 2023
1st Budapest – Cluj Meeting in Singularities and Beyond, Cluj-Napoca, Romania

Signatures of Milnor fibers 2023
Algebraic Geometry and Differential Topology Seminar, Rényi Institute, Budapest, Hungary

Feats and mysteries of holomorphic map germs (video) 2022
Algebraic Geometry and Differential Topology Seminar, Rényi Institute, Budapest, Hungary

Monodromía de gérmenes de funciones analíticas sin puntos fijos 2021
IX Encuentro de jóvenes topólogos, Sevilla, Spain

Monodromy of germs of analytic functions without fixed points (video) 2021
Iberoamerican Webminar of Young Researchers in Singularity Theory and related topics , On-line

Monodromy of germs of analytic functions without fixed points (video) 2020
16th International Workshop on Real and Complex Singularities, São Carlos, Brazil

On Whitney equisingular unfoldings of corank 1 map germs 2020
V Congreso de Jóvenes Investigadores Castelló 2020, Castellón, Spain

Monodromy of germs of analytic functions without fixed points 2020
3rd BYMAT Conference (Bringing Young Mathematicians Together), Valencia, Spain

On Whitney equisingular unfoldings of corank 1 map germs 2019
6th International Workshop on Singularities in Generic Geometry and Applications, Valencia, Spain

(...)

A weak version of Mond's Conjecture for corank 1 maps and applications 2019
Congreso Bienal de la Real Sociedad Matemática Española 2019, Santander, Spain

(...)

Posters

A weak version of Mond's Conjecture for corank 1 maps and applications 2018
15th International Workshop on Real and Complex Singularities, São Carlos, Brazil

Other merits

Teaching Experience

Universitat de València: **180 h** 2018–2020
Valencia, Spain

Innovation in education article 2022
Comunidades de práctica predoctorales: Diez años del Seminari Predoc (w/ E. Cosme Llópez, et al.)
IN-RED 2022 https://doi.org/10.4995/INRED2022.2022.15977

Budapesti Műszaki és Gazdaságtudományi Egyetem: **28 h** 2023
Budapest, Hungary

Universidad Autónoma de Madrid: **192 h** 2025–2026
Madrid, Spain

Universidad Politécnica de Madrid: **40 h** 2026
Madrid, Spain

Research

Extracurricular Internship, *Ingenia-Telecom S.L.* (180h) 2015
Performing R&D in machine learning.

Collaboration Internship, Department of Geometry and Topology, *Universitat de València* 2015–2016
Learning Homology Theory and giving a seminar related to it. Supervisor: R. Oset Sinha.

Introduction to research grant, *Instituto de Ciencias Matemáticas* (Madrid) 2016
Functional analysis and quantum information theory. Supervisor: C. Palazuelo Cabezón.

Experience in organisation of events

6th International Workshop on Singularities in Generic Geometry and Applications 2019

IV SEMF's winter symposium 2019

VIII SEMF's summer school 2020

Predoc seminar in Universitat de València, Faculty of Mathematics 2019–2021

Dissemination work / Popular science

What does *Mathematics* mean? (in Spanish) 2016
Talk in Blasco Ibáñez Higschool, in Cullera (Valencia), Spain

***Una oca en mi café* Cycle of talks** (in Spanish) (videos) 2018
Organizer of this cycle of talks, Valencia, Spain

About brains and maths (in Spanish) (videos) 2018
In the cycle of talks *Una oca en mi café*, Valencia, Spain

Noosfera 47: Topología y matemáticas que no se desayunan (in Spanish) 2021
La Razón's podcast *Noosfera* (link), Spain

The Arrival 2025
Cine y ciencia, a talk after the sceening of a film, Madrid, Spain

Noosfera 303: La IA y... ¿el fin de la matemática? (in Spanish) 2026
Guest with A. Forte and M. Camarasa in *La Razón*'s podcast *Noosfera* (link), Spain